IEEE STD 830 / ISO/IEC/IEEE 29148 COMPLIANT

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

AI-Powered Risk & Relocation Decision Support System

Problem Statement ID: 26191

Intelligent Identification of Hazard-Based Red Zones, Carrying Capacity Assessment, and Immediate Relocation Needs for Vulnerable Habitations

Event: Smart India Hackathon 2026

Theme: Disaster Management

PS Category: Software

Team ID: 165338

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional and non-functional requirements for the **AI-Powered Risk & Relocation Decision Support System** developed for Smart India Hackathon 2026 under Problem Statement ID **26191**. The system is designed to help disaster management authorities identify hazardous habitations, evaluate the carrying capacity of nearby safe host sites, and prioritize vulnerable communities for timely pre-monsoon relocation.

1.2 Document Conventions

- **FR-X.X:** Denotes a specific Functional Requirement.
- **NFR-X.X:** Denotes a specific Non-Functional Requirement.
- **Priority Tiers:** Habitations are ranked into **Immediate (<30 days), High, Medium (1–6 months), and Low** relocation priority levels.
- **Spatial Resolution:** All terrain and risk mapping operates on a uniform **10m × 10m spatial grid** (9x sharper than standard 30m maps).

1.3 Intended Audience and Reading Suggestions

- **State & District Disaster Management Authorities (SDMAs / DDMAs):** Focus on Sections 1, 2, and 4 for the operational **Detect** → **Assess** → **Act** workflow.
- **District Magistrates (Collectors):** Focus on Sections 2.3, 4.5, and 5.2 covering SHAP explainability scores, priority lists, and final administrative sign-off.
- **Developers, GIS Engineers & Evaluators:** Focus on Sections 3, 4, 5, and 6 for the PostGIS database structure, FastAPI architecture, XGBoost + AHP hybrid model, and offline PWA specifications.

1.4 Product Scope

Traditional disaster relocation planning relies on manual field analysis, static maps, and disconnected departmental files, taking up to **3 months**. Our software automates this workflow to reduce relocation planning time from **3 months to 48 hours**:

- **Detect:** Combines **Sentinel-1 & 2** satellite data and **ISRO Bhuvan 10m DEM** terrain data with a hybrid **XGBoost + AHP** model (60% AI + 40% terrain rules) to predict habitation risk scores (0–100%) and delineate 4 risk zones on a 10m × 10m grid.
- **Assess:** Counts current population using census records + AI rooftop detection, searches within a **5–25 km radius** for safe sites (slope < 12°), and checks 5+ carrying capacity parameters against **NDMA relief norms** (3.5 sq.m land and 55L water/day per person).
- **Act:** Generates automated relocation priorities (Immediate, High, Medium, Low), displays transparent **SHAP** explanations on a React + Leaflet dashboard, supports offline field operations via **PWA (<50 MB PMTiles)**, and exports 1-click PDF/GeoJSON reports.

1.5 References

1. **ISRO Bhuvan (`bhuvan.nrsc.gov.in`):** 10m DEM slope maps and 10m × 10m spatial grid (9x sharper risk mapping).
2. **Copernicus Sentinel-1 & 2 (`dataspace.copernicus.eu`):** All-weather Synthetic Aperture Radar (works through heavy clouds) and terrain risk data.
3. **National Disaster Management Authority – NDMA Norms (`ndma.gov.in`):** Relief and carrying capacity norms of 3.5 sq.m land and 55L water/day per person.
4. **IMD & OpenStreetMap (`mausam.imd.gov.in`, `openstreetmap.org`):** Free rainfall APIs, roads, healthcare, schools, and building rooftops for 5+ capacity checks within 5–25 km.
5. **UN Sendai Framework & IEEE Geo-AI Literature:** Benchmarks for XGBoost + AHP (91.4% accuracy), 48-hour planning speed, 35–40% relief spend reduction, and 80% open-source tech savings.
6. **Smart India Hackathon 2026 Portal (`sih.gov.in`):** Problem Statement ID 26191 specifications.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The system is a 100% open-source web and offline-capable GIS decision support tool that eliminates proprietary software licensing costs (80% cheaper than ArcGIS) and incurs ₹0 data acquisition cost. It is designed to integrate with **36 SDMAs and 700+ DDMAs** across India via standardized **OGC REST APIs**.

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[ LAYER 1: DATA INPUTS ] Satellite / DEM (Sentinel-1/2, Bhuvan 10m) |
Census & AI Rooftops | OpenStreetMap | IMD Hazard Records ↓ [
LAYER 2: PROCESSING & ANALYSIS ] GIS + PostGIS Spatial Database |
XGBoost (60%) + AHP Rules (40%) + SHAP (4 Risk Zones Mapping | 5–25 km
Safe-Site Search | 5+ Parameter Capacity Check) ↓ [ LAYER 3:
BACKEND SERVICES ] Python FastAPI (Connects spatial data, AI analysis,
and OGC REST map services in <45s) Outputs: Spatial Database
(PostGIS) | Red Zone Map (≥0.8) | Ranked Safe Sites | Fit
Verification ↓ [ LAYER 4: AUTHORITY DASHBOARD ] React + Leaflet Web
App | Offline PWA (PMTiles <50 MB) | Priority Lists | 1-Click
PDF/GeoJSON Export
```

2.2 Product Functions

1. **GIS Data Preparation:** Aligns 10m DEM, village boundaries, population counts, and OSM infrastructure (rivers, roads, hospitals, schools) into a 10m × 10m grid.
2. **AI Risk Assessment:** Computes hazard and vulnerability risk scores (0–100%) using XGBoost + AHP and generates SHAP explanations.
3. **Safe Site Identification:** Searches within 5–25 km for safe areas with slope < 12° and verifies road, water, health, and school access.

4. **Carrying Capacity Assessment:** Enforces NDMA land (3.5 sq.m/person) and water (55L/day/person) norms, computes available capacity (`Safe - Existing`), and recommends Single or Split sites.
5. **Relocation Priority & Dashboard:** Categorizes habitations into Immediate, High, Medium, and Low priorities on an interactive React + Leaflet map with PDF/GeoJSON export.

2.3 User Classes and Characteristics

User Class	Role Description	Primary System Interactions
District Magistrate (DM / Collector)	Chief district authority responsible for approving resettlements	Reviews transparent SHAP risk scores, inspects Immediate/High priority queues, and provides final administrative sign-off
SDMA / DDMA Planners	State and District disaster management officers	Monitors 4 risk zones (Red Zone ≥ 0.8), evaluates Single vs. Split host site capacity fits, and exports GeoJSON/PDF plans
Field Surveyors	Ground inspection teams operating in remote rural blocks	Uses PWA offline mode (<50 MB PMTiles) without internet connectivity and downloads 1-click PDF verification sheets

2.4 Operating Environment

- **Client Side:** Desktop and mobile browsers supporting Progressive Web Apps (PWA), Service Workers, Leaflet.js, and **PMTiles (<50 MB)** offline vector/raster tiles.
- **Server Side:** Python **FastAPI** backend running **XGBoost, SHAP, GDAL/QGIS** processing pipelines connected to a **PostgreSQL + PostGIS** spatial database.

2.5 Design and Implementation Constraints

- **Zero License & Data Cost:** Must exclusively use open-source libraries and free APIs (ISRO Bhuvan, Sentinel, IMD, OSM).
- **NDMA Policy Compliance:** Carrying capacity calculations must enforce ≥3.5 sq.m land and ≥55L water/day per person and require final DM approval.
- **OGC Standards:** Map layers and exports must conform to OGC REST API and GeoJSON standards for compatibility with state portals.

2.6 Assumptions and Dependencies

- Sentinel-1 radar imagery provides uninterrupted terrain observations during heavy monsoon cloud cover.
- In regions with sparse past disaster records, the 40% AHP terrain rule component ensures stable risk classification.

3. EXTERNAL INTERFACE REQUIREMENTS

3.1 User Interfaces

- **Interactive GIS Map Panel (React + Leaflet):** Displays 4 risk zones on a 10m × 10m grid, highlighting Red Zones (score ≥ 0.8), vulnerable villages, and ranked safe host sites within 5–25 km.
- **SHAP Explainability Card:** Shows the 0–100% risk score of any clicked habitation along with the exact contribution of slope, rainfall, and terrain factors.
- **Carrying Capacity & Fit Verification Panel:** Displays host site slope (<12°), land capacity, water capacity, road/school/hospital proximity, and Single/Split site allocation details.
- **Action & Export Toolbar:** Provides tabs for Immediate (<30 days), High, Medium (1–6 months), and Low priority habitations, DM sign-off toggle, and 1-click PDF/GeoJSON download buttons.

3.2 Hardware Interfaces

The system runs on standard laptops, tablets, and smartphones without specialized hardware. Local device storage is utilized via browser cache to store offline PMTiles map packages (<50 MB).

3.3 Software & API Interfaces

- **ISRO Bhuvan & Sentinel-1/2 Interface:** Imports 10m DEM rasters and all-weather radar terrain layers.
- **IMD & OpenStreetMap (OSM) Interface:** Fetches rainfall feeds and road, river, healthcare, school, and rooftop vector layers.
- **PostGIS Interface:** Executes spatial buffer queries (5–25 km), slope filtering (<12°), and grid intersection math.
- **OGC REST API Interface:** Connects system outputs to 36 SDMAs and 700+ DDMAs.

3.4 Communications Interfaces

Uses HTTPS REST APIs exchanging JSON and GeoJSON payloads, paired with PWA Service Workers for offline caching in zero-connectivity zones.

4. SYSTEM FEATURES (FUNCTIONAL REQUIREMENTS)

4.1 Feature 1: GIS Data Preparation & AI Rooftop Counting

- **FR-1.1:** The system shall ingest **ISRO Bhuvan 10m DEM** data and generate uniform **10m × 10m** spatial grid cells (9x sharper than 30m maps).
- **FR-1.2:** The system shall process **Sentinel-1 radar** imagery to detect terrain risk through heavy monsoon clouds.
- **FR-1.3:** The system shall import rivers, roads, hospitals, and schools from **OpenStreetMap (OSM)** and rainfall data from **IMD**.
- **FR-1.4:** To compensate for old census records, the system shall combine census baselines with **AI rooftop counting** from satellite imagery to estimate current village population.

4.2 Feature 2: Hybrid AI Risk Assessment & Explainability

- **FR-2.1:** The system shall predict a habitation risk score (0–100%) using a hybrid model combining **60% XGBoost** predictions and **40% AHP terrain/slope rules**.
- **FR-2.2:** The system shall map **4 risk zones** and classify grid cells with a risk score **≥ 0.8 (80%)** as **Red Zones**.
- **FR-2.3:** The system shall compute **SHAP (Explainable AI)** scores for each habitation to explain the exact reasons behind its risk score to authorities.

4.3 Feature 3: Safe Host Site Identification

- **FR-3.1:** For every habitation requiring relocation, the system shall search for safe candidate sites within a **5 km to 25 km radius** so villagers stay near their farms and jobs.
- **FR-3.2:** The system shall filter candidate sites to retain only areas where the terrain slope is **less than 12° (slope < 12°)**.
- **FR-3.3:** The system shall verify access to roads, water sources, healthcare facilities, and schools for each candidate site.

4.4 Feature 4: Carrying Capacity Assessment & Single/Split Matching

- **FR-4.1:** The system shall analyze **5+ carrying capacity parameters**: population, water, healthcare, roads, schools, and buildable land.
- **FR-4.2:** The system shall enforce **NDMA relief norms** requiring a minimum of **3.5 sq.m of land per person** and **55 liters of water per person per day**.
- **FR-4.3:** The system shall calculate net available capacity as: `Available Capacity = Safe Capacity - Existing Population`.
- **FR-4.4:** The system shall automatically recommend either a **Single Site** or a **Split Site** (dividing relocated households across two or more nearby safe sites if a single site lacks sufficient capacity).

4.5 Feature 5: Relocation Priority & Authority Dashboard

- **FR-5.1:** The system shall generate automated relocation priority queues categorized into **Immediate (<30 days), High, Medium (1–6 months), and Low** priority.
- **FR-5.2:** The system shall require final approval sign-off by the **District Magistrate (DM / Collector)** before finalizing a relocation plan.
- **FR-5.3:** The system shall provide **1-click PDF and GeoJSON export** of risk maps, ranked safe sites, and priority lists.

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance Requirements

- **NFR-1.1 (Processing & Response Time):** Backend risk and capacity calculations shall execute in **under 45 seconds (<45s)**, meeting the overall target response time of **<5 minutes** after new data arrives.
- **NFR-1.2 (Planning Speed):** The platform shall reduce end-to-end relocation planning time from **3 months to 48 hours**.
- **NFR-1.3 (Model Accuracy):** The hybrid XGBoost + AHP model shall achieve a target accuracy of **>90%** (benchmarked at **91.4% accuracy**).

5.2 Safety & Policy Compliance

- **NFR-2.1 (NDMA & Sendai Alignment):** All outputs shall comply with NDMA relief standards (3.5 sq.m land, 55L water/day) and the UN Sendai Framework.
- **NFR-2.2 (Overcrowding & Construction Freeze):** The system shall prevent host-site overcrowding through strict capacity caps and support a Red-Zone freeze to block risky new construction.

5.3 Security & Privacy Requirements

- **NFR-3.1 (Role-Based Approval):** Final relocation sign-off privileges shall be restricted to authenticated District Collectors / Magistrates.
- **NFR-3.2 (Privacy Preservation):** Demographic calculations shall only use aggregate census numbers and satellite rooftop counts without storing personal citizen identities.

5.4 Software Quality & Viability Attributes

- **NFR-4.1 (Offline Field Capability):** The PWA shall package district maps into **PMTiles under 50 MB (<50 MB)** for seamless offline operation in remote areas.
- **NFR-4.2 (Scalability):** OGC REST API architecture shall support integration across **36 SDMAs and 700+ DDMAs**.
- **NFR-4.3 (Economic Viability):** Built on a 100% open-source stack (80% cheaper than ArcGIS) with ₹0 data cost, cutting post-disaster relief spending by **35–40%**.

6. APPENDIX & GLOSSARY

Acronym / Term	Definition
DEM	Digital Elevation Model – 3D slope map (10m resolution from ISRO Bhuvan)
AHP	Analytic Hierarchy Process – Rule-based scoring method (provides 40% terrain rules alongside 60% AI)
SHAP	SHapley Additive exPlanations – Explainable AI score showing why an area is risky
PMTiles	Protomaps Tiles – Single-file offline map format keeping district maps under 50 MB
PWA	Progressive Web App – App that runs inside a browser and works offline in the field
OGC	Open Geospatial Consortium – Global standard for GIS REST APIs to plug into state portals
SDMA / DDMA	State / District Disaster Management Authority
IMD & OSM	India Meteorological Department (weather) & OpenStreetMap (roads, schools, hospitals, rooftops)
DM	District Magistrate (District Collector) – Final authority approving the relocation plan